

In this supplementary file we include the following information:

Figures S1 to S19

Tables S1 to S7

Section 1

1 INTRODUCTION

We include supplementary figures of the study area surface elevation, sample Sentinel 1 VV and VH time series, glacier hypsometry, crossover analysis for the airborne radar data for each year and compared to the surface-coupled ground penetrating radar in 2023, SWE values per elevation bin when including interpolation to Mt. Logan cores, SWE values per elevation bin without interpolation, the winter input mass by survey year, two images of a snow pit wall, a plot of the snow density against depth, weather station time series plots of temperature and precipitation, Sentinel 1 VV timeseries near the 2018 Kaskawulsh Core, and the radar amplitude for the last summer surface from each survey.

Data tables provide information regarding the individual snow pits and cores data as well as individual hyperbolic analysis surveys. We also include tables containing the values from plots elsewhere in this work regarding snowline/ELA elevation, glacier average SWE, and the winter mass balance.

Lastly, we include a section that discusses the propagation of uncertainty for calculations of snow depth and real permittivity from two way travel time, the uncertainty for the 50 m elevation bins, and the uncertainty for the glacier average SWE and winter mass balance.

19 2 SUPPLEMENTARY FIGURES

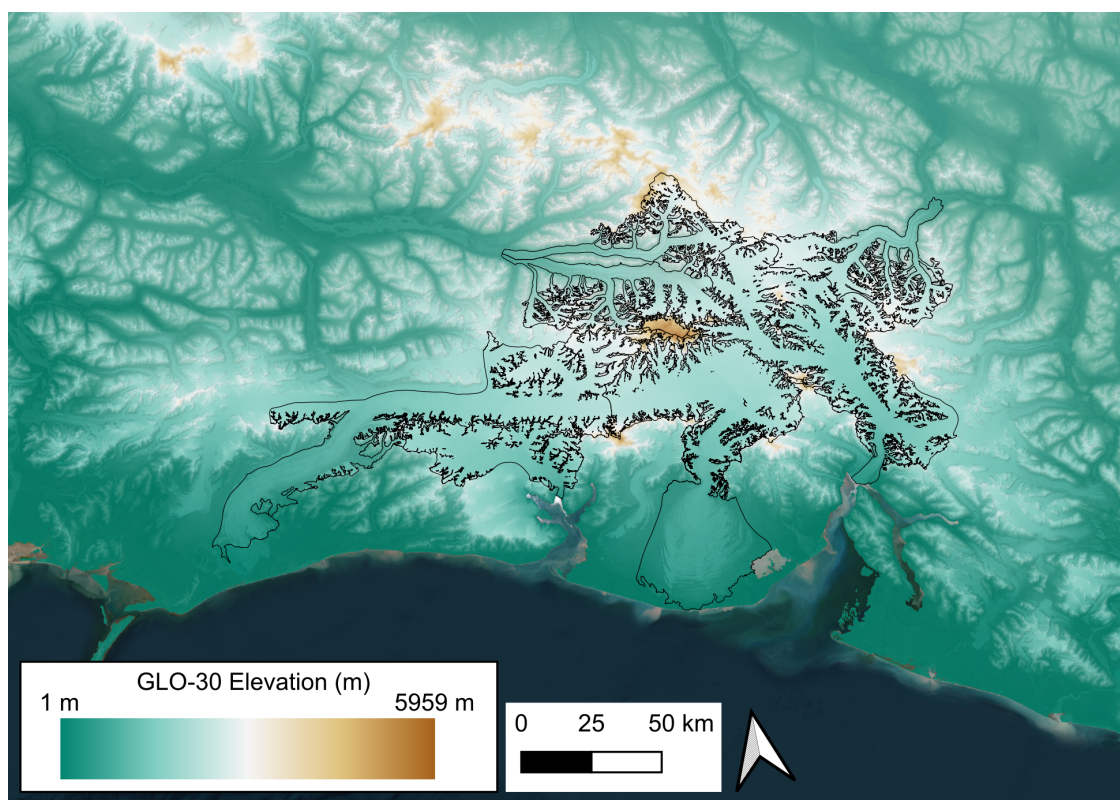


Fig. S1. Surface elevation of the St. Elias Mountains from the Copernicus GLO-30 digital elevation model. The glaciers in our study are outlined in black. Elevations reach a maximum of 5959 m a.s.l. at the summit of Mt. Logan. Data from OpenTopography (2021).

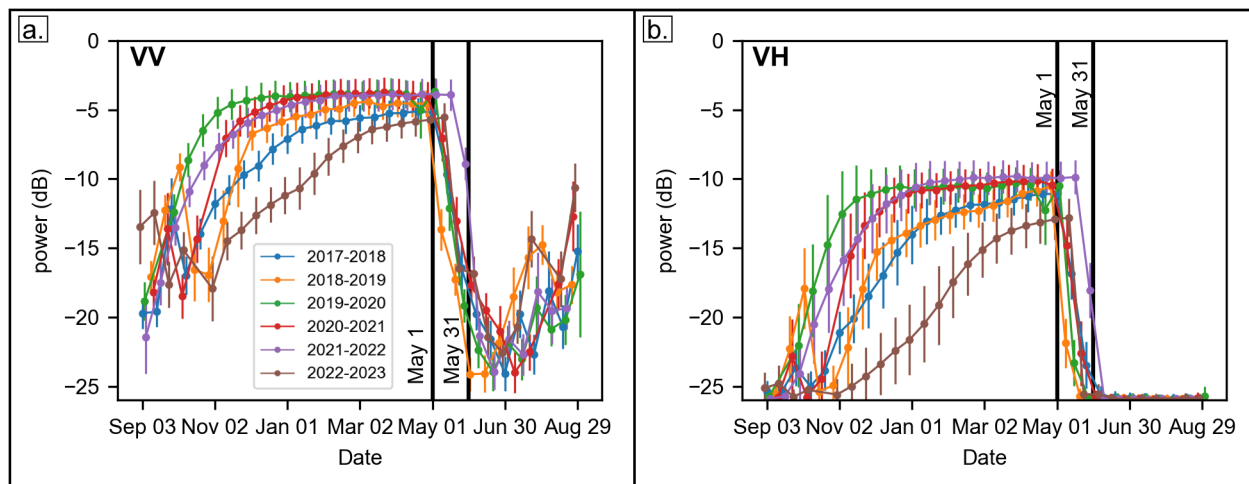


Fig. S2. Time series of Sentinel 1 (a) VV and (b) VH reflectivity at the 2023 Seward core site. Error bars are one standard deviation of the power in the sampling area. Similar features are visible for both polarizations, but with a lower returned power by ~ 6 dB for VH. While the features in May were observable in both polarizations, features in the early fall are less prominent in VH. Additionally, the VH polarization dips below the noise equivalent sigma zero (-26 dB) for much of the summer, whereas the VV polarization does not.

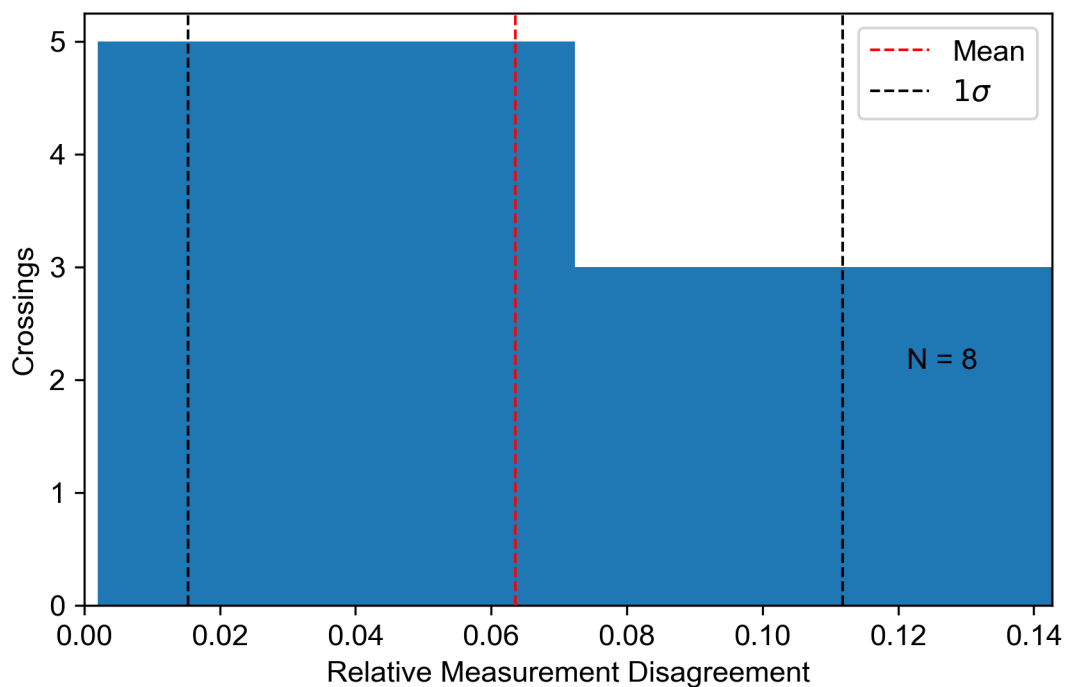


Fig. S3. Spread of the 2018 crossover points in the relative measurement space.

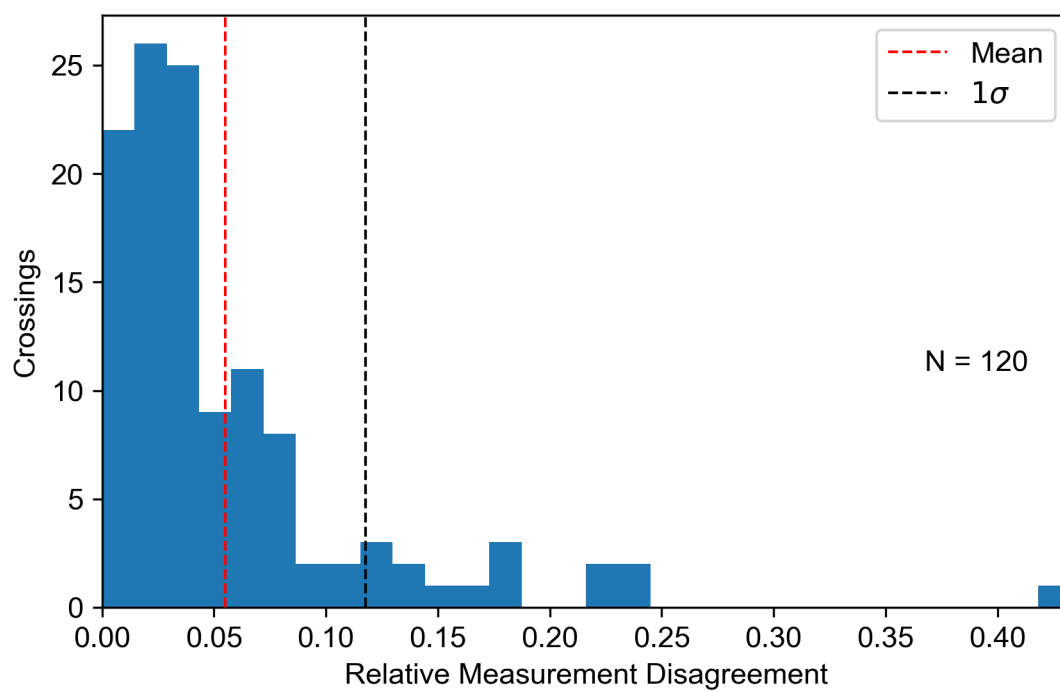


Fig. S4. Spread of the 2021 crossover points in the relative measurement space. Two thirds of the 2023 crossover points fall below the mean.

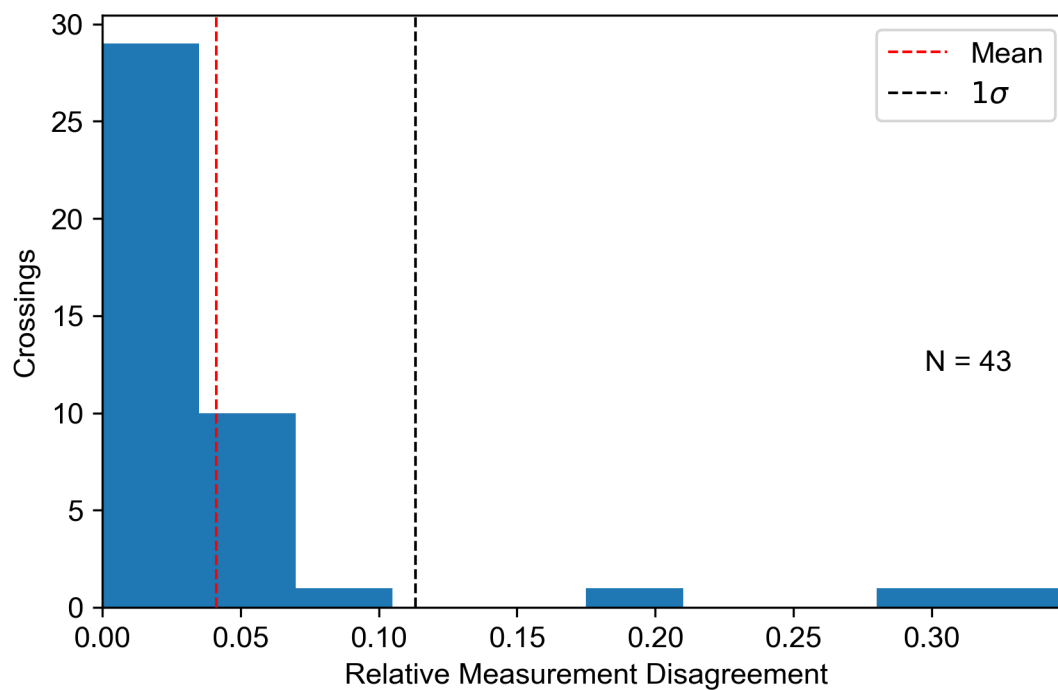


Fig. S5. Spread of the 2023 crossover points in the relative measurement space. Two thirds of the 2023 crossover points fall below the mean. The extremely different points tend to be in areas where tracking reflectors for long stretches was challenging or impossible. Thus, large differences are likely due to the picking the shallower reflector in radargram and the last summer surface in the other due to the similarities when both reflectors were not present.

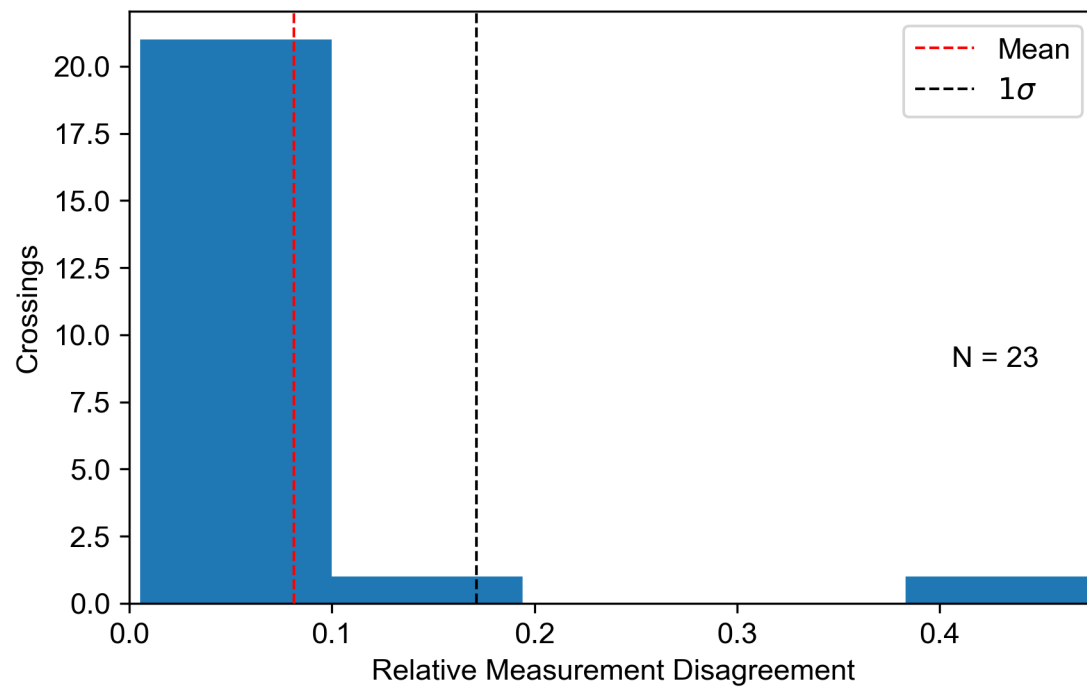


Fig. S6. Spread of the 2023 crossover points between the surface GPR and the airborne radar. At all but two crossover points, the airborne radar was deeper than the surface GPR.

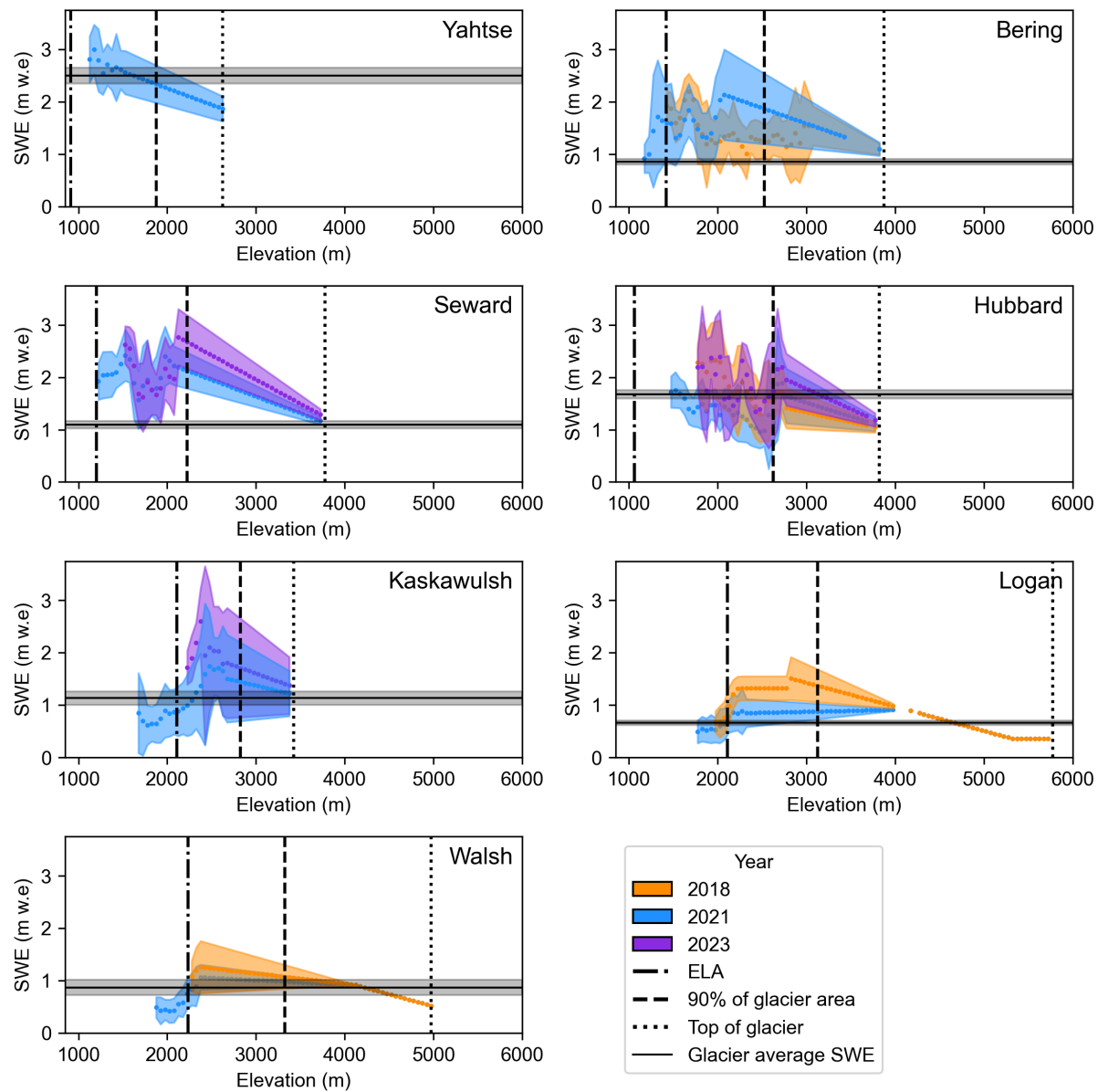


Fig. S7. SWE values for each elevation bin including the interpolation to the high elevation core sites on Mt. Logan. The glacier average SWE distributes the input mass to the accumulation area over the total glacier area, which causes the this value to be lower than most elevation bin values.

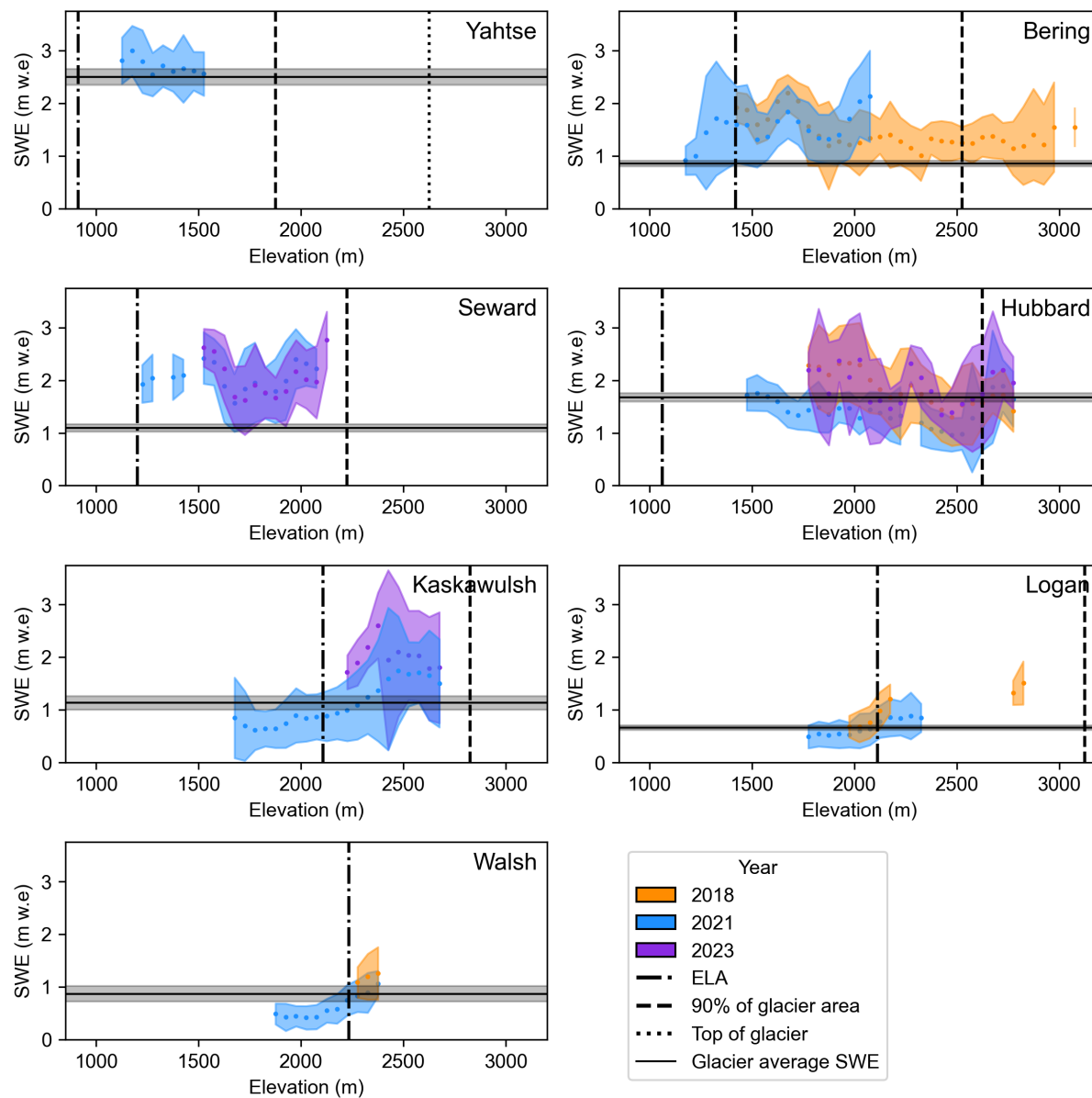


Fig. S8. SWE values for each elevation bin without the interpolation of missing data within elevations sampled by the airborne radar. Interpolation within the radar data only needed to be done of the Seward in 2021, Hubbard in 2021, and Logan in 2018. The glacier average SWE distributes the input mass to the accumulation area over the total glacier area, which causes the this value to be lower than most elevation bin values.

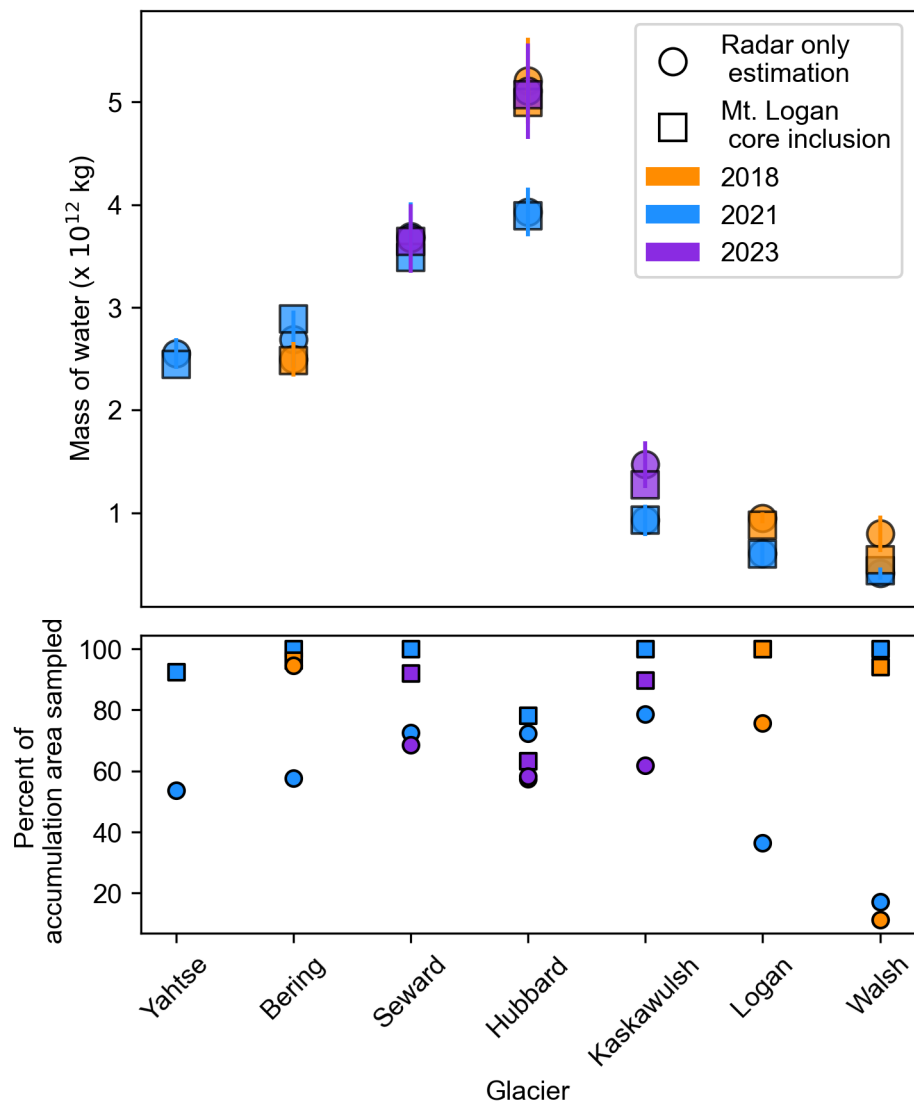


Fig. S9. (a) Winter mass balance in mass of water for each glacier by survey year and (b) corresponding percent of the accumulation area sampled. Glaciers are ordered roughly by their distance from the coast, closer (left) to further (right). Here we only show the uncertainties from the radar only estimation. In this plot we show only the 2σ uncertainty for the radar only estimation.

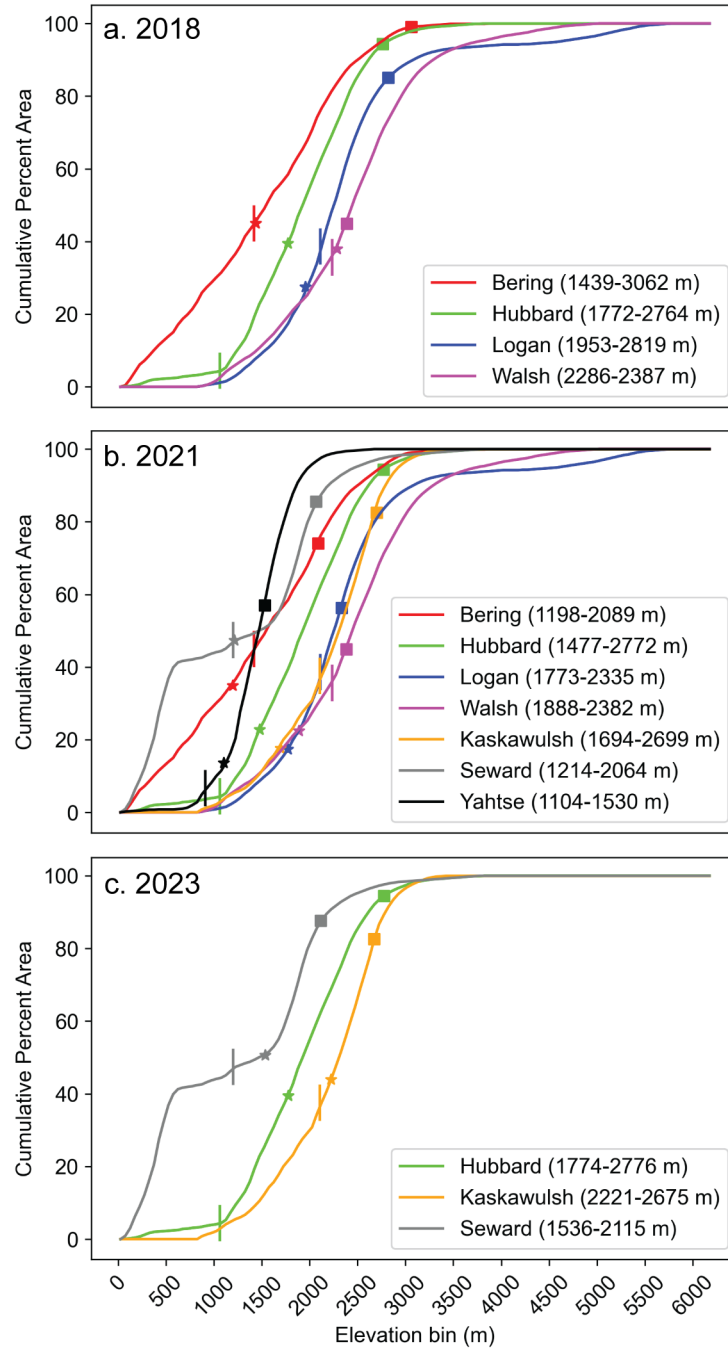


Fig. S10. Cumulative percent of the glacier area at each elevation bin for (a) 2018, (b) 2021, (c) 2023. Each glacier is marked by the lowest (star) and highest (square) surface elevation with airborne data and the mean ELA (vertical line). Glacier labels in the legend are the surface elevation range for the airborne data. Relevant numerical values are listed in Table S4.

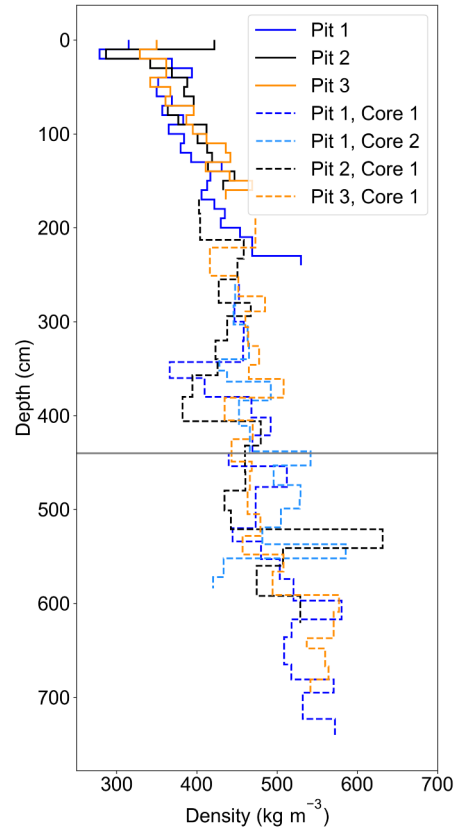


Fig. S11. Measured densities for the pits (solid lines) and cores (dashed). We omit the 9% uncertainty on each measurement in the plot for ease of seeing the measured value. Discontinuities at the pit to core interface are attributed to the change in sampling techniques. The gray horizontal line marks the approximate last summer surface depth.

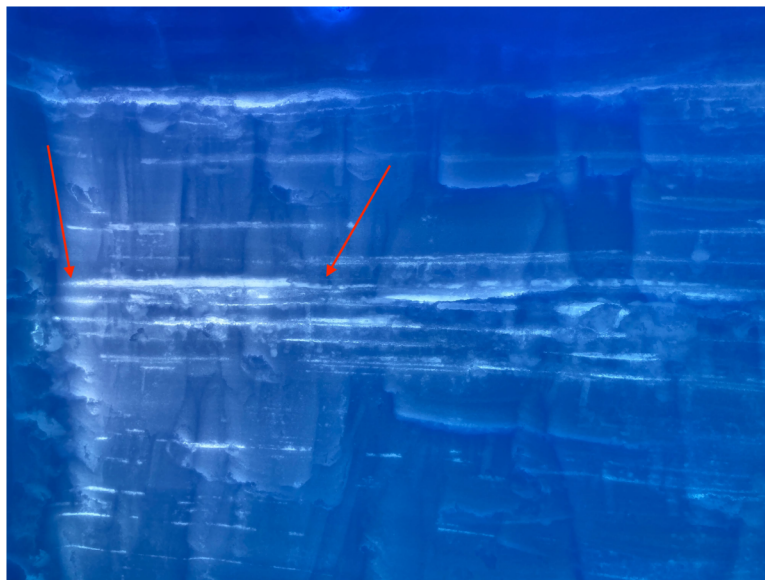


Fig. S12. Upper section of Snow Pit 1. Photo was taken by digging two adjacent snow pits with a preserved snow wall between them. The blue color is due to a blue tarp covering the snow pit. The ice layers were much more apparent through the snow wall than looking at the wall of a single snow pit. Red arrows in Figs. S12 and S13 are the same ice lens.

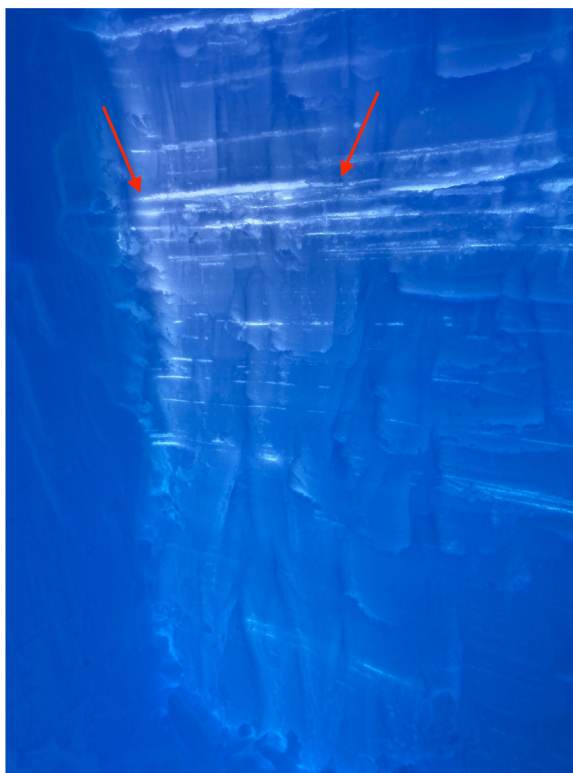


Fig. S13. Deeper section of Snow Pit 1, with the snow pit bottom in the lower left. After the concentration of ice layers shown in Fig. S12, they become less common deeper in the snow column.

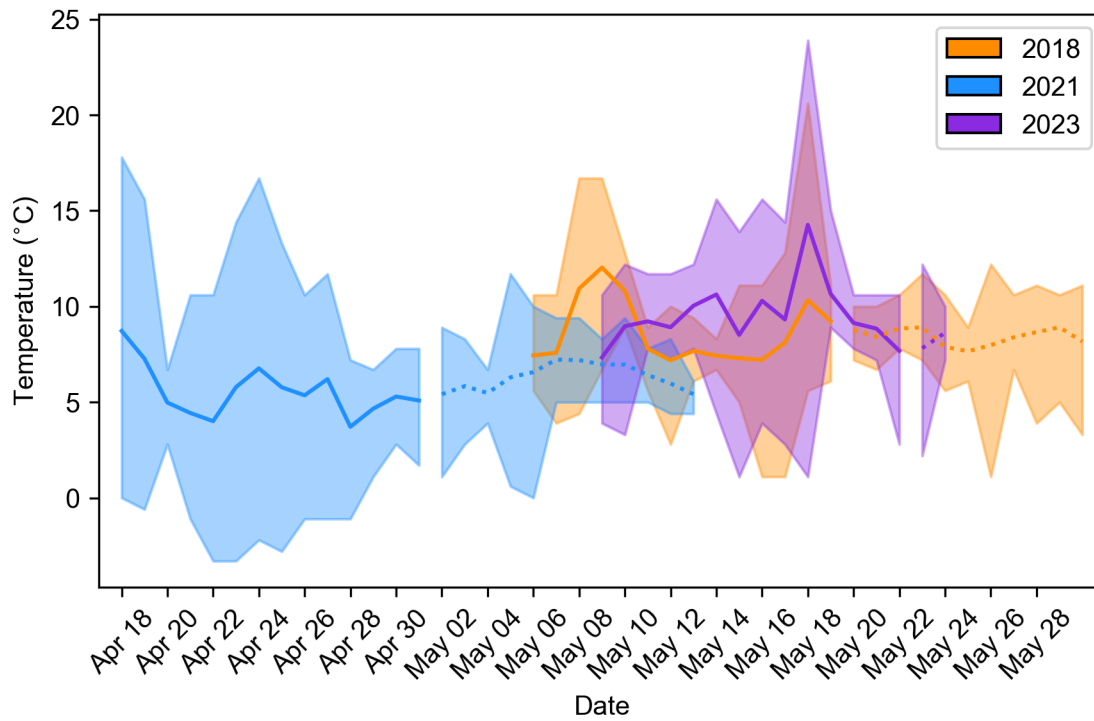


Fig. S14. Daily temperature time series for Yakutat Airport (10 m a.s.l.). Solid line is the daily mean, with the dotted lines being the daily mean during airborne data acquisition. Shading ranges between the daily minimum and maximum temperature.

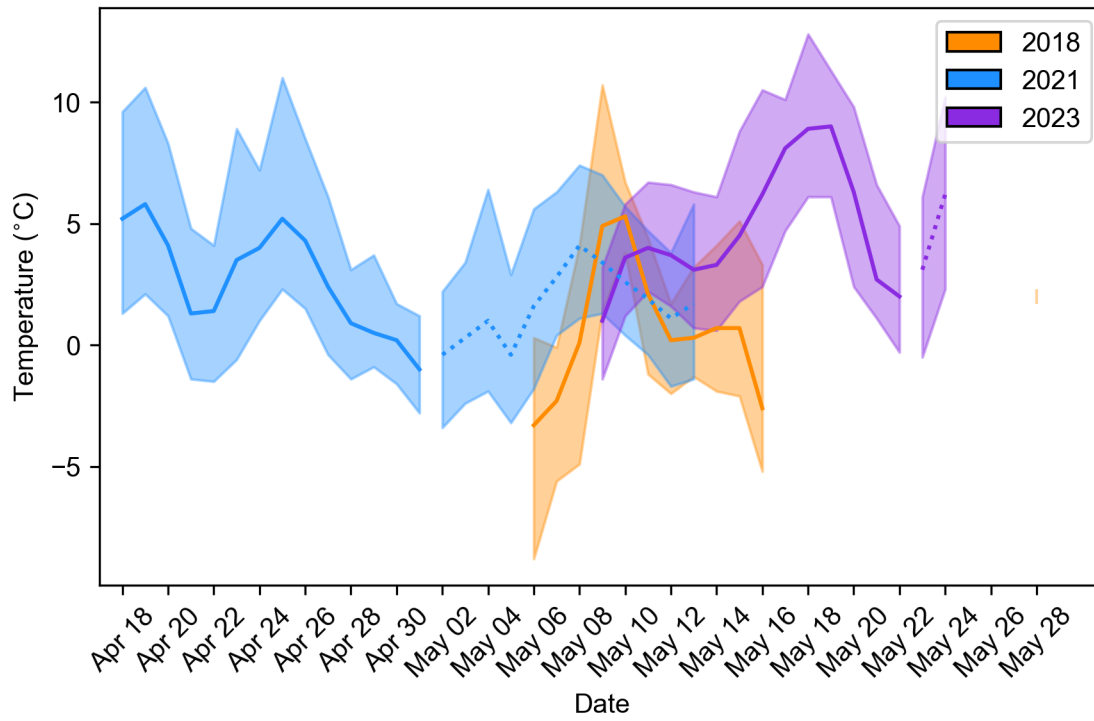


Fig. S15. Daily temperature time series for Tana Knob (1050 m a.s.l.). Solid line is the daily mean, with the dotted lines being the daily mean during airborne data acquisition. Shading ranges between the daily minimum and maximum temperature. Tana Knob was missing data in 2018, which is why there is less data coverage than compared to Yakutat (Fig. S14).

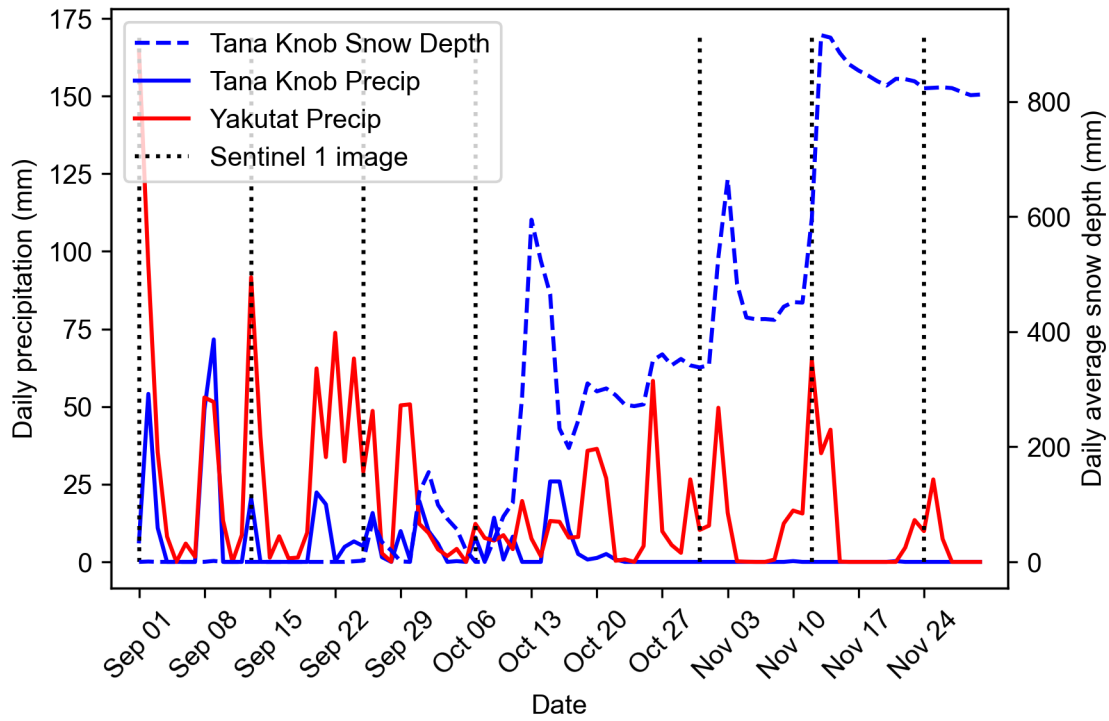


Fig. S16. Daily precipitation total (solid lines) for Tana Knob and Yakutat and the daily mean snow depth (dashed line) at Tana Knob for September 1st through November 30th, 2022. The lack of precipitation at Tana Knob after mid-October is likely due to no melt or liquid precipitation. The vertical dotted lines mark days when Sentinel 1 imagery was collected for the selected path and frame. There was no mid-October, 2022 image for the selected path and frame.

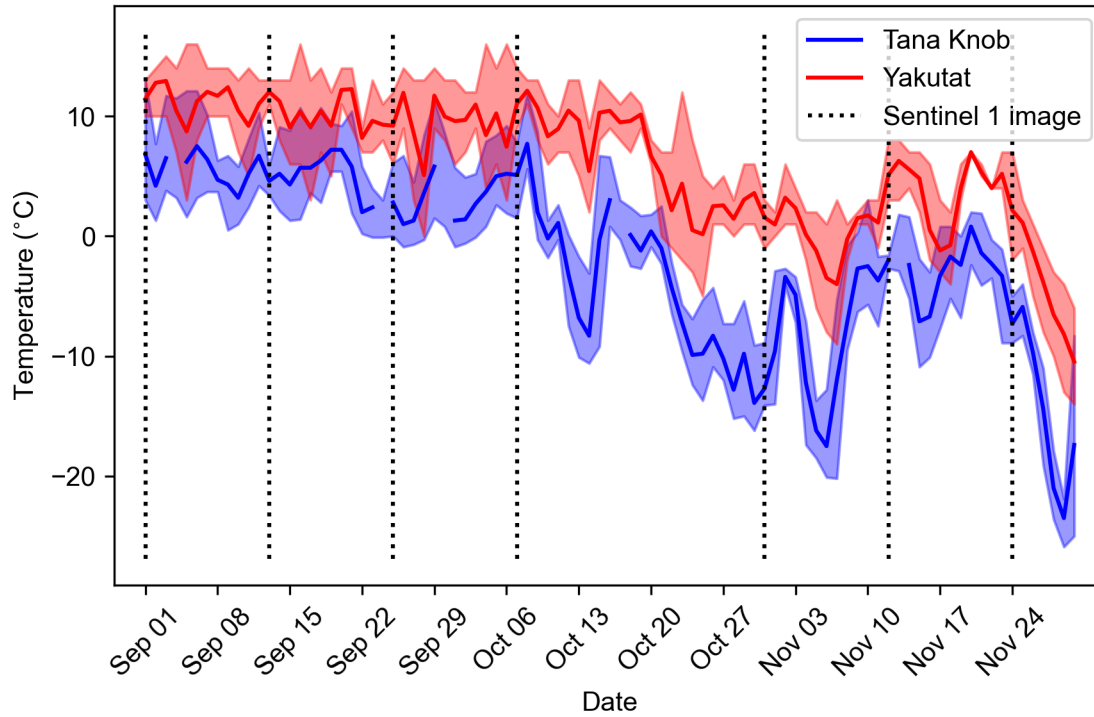


Fig. S17. Daily average temperature for Tana Knob and Yakutat for September 1st through November 30th, 2022. Shading ranges between the daily minimum and maximum. The vertical dotted lines mark days when Sentinel 1 imagery was collected for the selected path and frame. There was no mid-October, 2022 image for the selected path and frame.

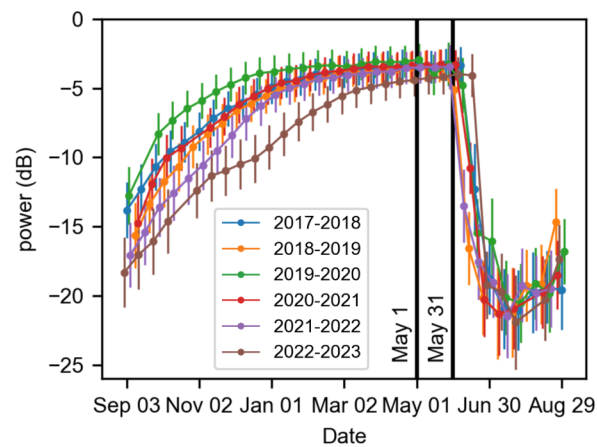


Fig. S18. Time series of Sentinel 1 reflectivity at the Kaskawulsh core site. Error bars are the one standard deviation of the power in the sampling area. Red line is the 2020-2021 accumulation season. No drop in reflectivity during fall 2020 is observed, which supports the interpretation of a strong, end of season ice layer that is not overlain by a fall rain on snow event. Also, the reflectivity drop from the onset of summer melting occurs later in the year than at the Seward sites due to the Kaskawulsh core site being at ~ 2650 m a.s.l., a kilometer higher than the Seward core sites.

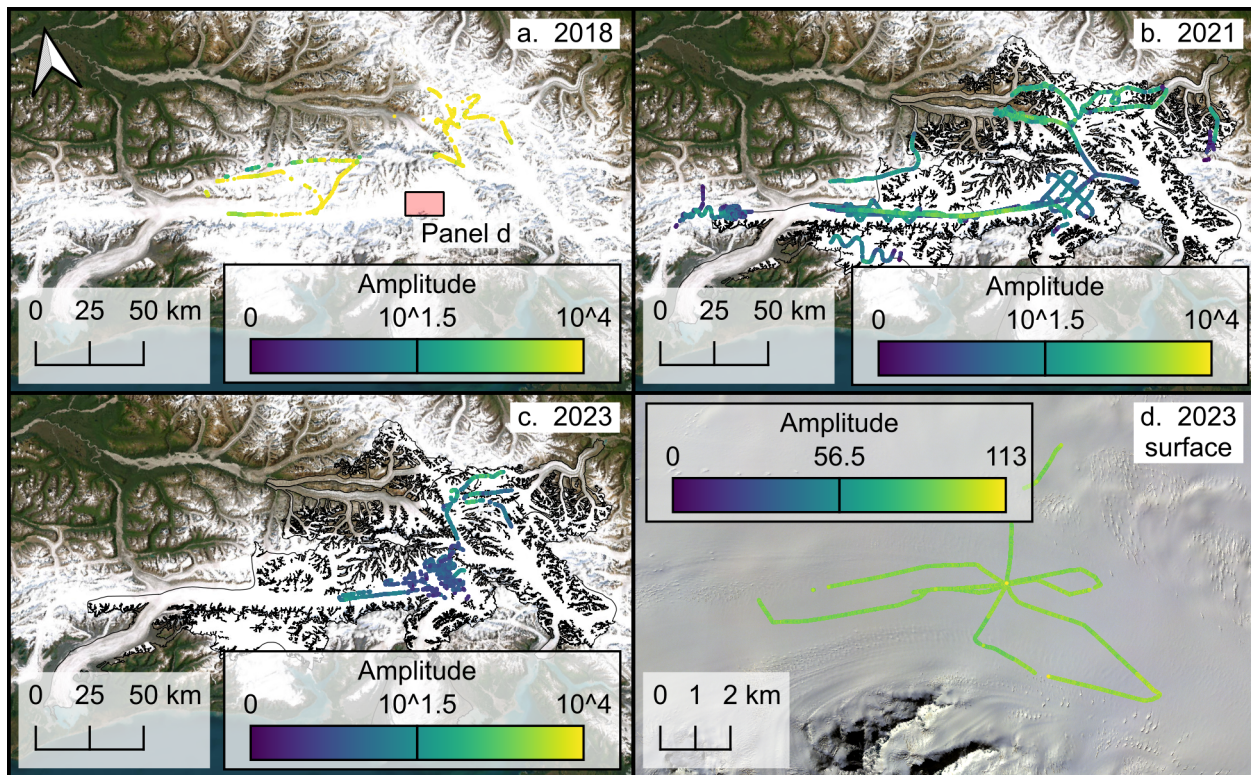


Fig. S19. Radar amplitude of the last summer surface for (a-c) the airborne radar and the (d) surface GPR. The scale for (a-c) the airborne radar is a logarithmic scale while the scale for the (d) surface GPR is a linear scale.

20 **3 SUPPLEMENTARY TABLES**

Table S1. Summary of snow pit and core collection date, pit depth, and core depth. Core depths are inclusive of the pit depths. Visualization of snow pit locations in Fig. 2.

Collection date	Latitude ($^{\circ}$ N)	Longitude ($^{\circ}$ W)	Pit	Core	Pit Max Depth (m)	Core Max Depth (m)
May 15, 2023	60.381139 N	140.277101 W	1	1	2.4	7.41
May 15, 2023	60.381139 N	140.277101 W	1	2	2.4	5.84
May 15, 2023	60.387577 N	140.292982 W	2	1	1.6	6.21
May 17, 2023	60.375174 N	140.245856 W	3	1	1.7	6.95

Table S2. SWE values and mass in gigatons (Gt) of water for each glacier using the radar only estimation and the sensitivity test including the interpolation to the Mt. Logan cores. Values are plotted in Figs. 8, 9, & S9. Uncertainties are reported as 2σ .

Glacier	Year	SWE (m w.e.); Radar	Gt (or 10^{12} kg) water;	SWE (m w.e.);	(or 10^{12} kg) water;
		only estimation	Radar only estimation	Mt. Logan core inclusion	Mt. Logan core inclusion
Hubbard	2023	1.80 ± 0.17	5.11 ± 0.46	1.79 ± 0.16	5.07 ± 0.43
Kaskawulsh	2023	1.40 ± 0.21	1.47 ± 0.23	1.20 ± 0.16	1.27 ± 0.17
Seward	2023	1.09 ± 0.10	3.67 ± 0.34	1.09 ± 0.08	3.64 ± 0.25
Bering	2021	0.88 ± 0.09	2.69 ± 0.28	0.95 ± 0.07	2.89 ± 0.21
Hubbard	2021	1.38 ± 0.08	3.93 ± 0.24	1.37 ± 0.08	3.89 ± 0.22
Kaskawulsh	2021	0.88 ± 0.15	0.93 ± 0.15	0.88 ± 0.12	0.93 ± 0.13
Logan	2021	0.52 ± 0.08	0.60 ± 0.11	0.51 ± 0.04	0.60 ± 0.05
Seward	2021	1.10 ± 0.10	3.69 ± 0.33	1.04 ± 0.07	3.49 ± 0.25
Walsh	2021	0.59 ± 0.12	0.41 ± 0.06	0.62 ± 0.03	0.44 ± 0.02
Yahtse	2021	2.50 ± 0.15	2.55 ± 0.15	2.40 ± 0.10	2.44 ± 0.11
Bering	2018	0.82 ± 0.05	2.49 ± 0.17	0.82 ± 0.05	2.48 ± 0.17
Hubbard	2018	1.85 ± 0.14	5.21 ± 0.42	1.77 ± 0.14	4.99 ± 0.38
Logan	2018	0.81 ± 0.04	0.95 ± 0.05	0.75 ± 0.03	0.88 ± 0.04
Walsh	2018	1.15 ± 0.27	0.80 ± 0.18	0.75 ± 0.05	0.54 ± 0.04

Table S3. Mean SWE values and mean mass in gigatons (Gt) of water for each glacier using the radar only estimation and the sensitivity test including the interpolation to the Mt. Logan cores. Winter input mass values are plotted in Fig. 9. Uncertainties are reported as 2σ .

Glacier	SWE (m w.e.); Radar	Gt (or 10^{12} kg) water;	SWE (m w.e.);	(or 10^{12} kg) water;
	only estimation	Radar only estimation	Mt. Logan core inclusion	Mt. Logan core inclusion
Bering	0.85 ± 0.05	2.59 ± 0.16	0.89 ± 0.04	2.69 ± 0.13
Hubbard	1.67 ± 0.08	4.74 ± 0.22	1.64 ± 0.08	4.65 ± 0.20
Kaskawulsh	1.13 ± 0.13	1.20 ± 0.14	1.04 ± 0.10	1.1 ± 0.11
Logan	0.66 ± 0.05	0.78 ± 0.06	0.63 ± 0.03	0.74 ± 0.03
Seward	1.10 ± 0.07	3.68 ± 0.24	1.06 ± 0.05	3.56 ± 0.18
Walsh	0.87 ± 0.15	0.60 ± 0.09	0.69 ± 0.03	0.49 ± 0.02
Yahtse	2.50 ± 0.15	2.55 ± 0.15	2.40 ± 0.10	2.44 ± 0.11

Table S4. For each glacier and survey year the percent of the accumulation area sampled by the airborne radar, and percent of the accumulation area that is above the lowest surface elevation with airborne radar data, the total area below, within, and above the airborne radar. Total areas reference the total glacier (accumulation and ablation) area. Visual representation of this data is in Fig. S10.

Glacier	Year	Percent of							
		Percent of	accumulation	Total	Total	Total	Total	Total	Total
		accumulation	area	area	area	area	area	area	area
		area	above	below	below	sampled by	sampled by	area above	area above
		sampled by radar	the lowest elevation with radar	radar (km ²)	radar (%)	radar (km ²)	radar (%)	radar (km ²)	radar (%)
Hubbard	2023	58.37	63.28	1119.62	39.5	1581.65	55.8	133.22	4.7
Kaskawulsh	2023	61.86	89.74	463.64	44	406.74	38.6	183.35	17.4
Seward	2023	68.63	92.02	1735.13	51.6	1213.92	36.1	413.61	12.3
Bering	2021	57.64	100	1107.19	36.6	1213.07	40.1	704.85	23.3
Hubbard	2021	72.28	78.14	717.13	25.3	1958.63	69.1	158.73	5.6
Kaskawulsh	2021	78.69	100	210.74	20	702.83	66.7	140.14	13.3
Logan	2021	36.48	100	204.84	17.4	513.28	43.6	459.13	39
Seward	2021	72.62	100	1593.9	47.4	1284.54	38.2	484.22	14.4
Walsh	2021	17.07	100	170.98	23.8	177.45	24.7	369.98	51.5
Yahtse	2021	53.64	92.51	138.6	13.6	510.57	50.1	369.93	36.3
Bering	2018	94.55	96.1	1424.83	47.1	1573.06	52	27.23	0.9
Hubbard	2018	57.43	63.28	1119.62	39.5	1556.14	54.9	158.73	5.6
Logan	2018	75.73	100	323.74	27.5	678.09	57.6	175.41	14.9
Walsh	2018	11.27	94.2	298.14	41.5	50.29	7	369.98	51.5

Table S5. Hyperbolic analysis results for the 1 m deep probe in Snow Pit 1. Data were collected on May 20, 2023.

Track	Time window (ns)	TWTT (ns)	Relative permittivity
44	500	12.5	3.51
45	100	12.744	3.65
46	100	12.363	3.43
47	100	12.461	3.49
Mean	—	12.517	3.52
TWTT stdev	—	0.162	—

Table S6. Hyperbolic analysis results for the 1.5 m deep probe in Snow Pit 1. Data were collected on May 20, 2023.

Track	Time window (ns)	TWTT (ns)	Relative permittivity
48	500	16.653	2.74
49	100	16.65	2.77
Mean	–	16.607	2.75
TWTT stdev	–	0.062	–

Table S7. Hyperbolic analysis results for the 1.2 m deep probe in Snow Pit 3. Data were collected on May 17, 2023.

Track	Time window (ns)	TWTT (ns)	Relative permittivity
32	500	13.477	2.83
33	500	13.232	2.73
Mean	–	13.355	2.78
TWTT stdev	–	0.17	–

Table S8. Data for the mean, median, standard deviation, in addition to the accumulation area ratio and total glacier area. The mean snowline is used as the ELA.

Glacier	Mean snowline (m)	Median snowline (m)	stdev snowline (m)	Accumulation area ratio	Total glacier area (km ²)
Bering	1421	1429	41	0.55	3025
Hubbard	1061	1064	24	0.96	2834
Kaskawulsh	2106	2140	100	0.62	1054
Logan	2112	2127	45	0.61	1177
Seward	1201	1209	102	0.53	3363
Walsh	2234	2247	57	0.62	718
Yahtse	912	874	91	0.93	1019

4 UNCERTAINTY PROPAGATION

4.1 Converting from two way travel time to other variables

Equations 1–4 were used to convert two way travel times to real permittivity of layer depth. Real permittivity (Eq. 1) and associated uncertainty (Eq. 2) when two way travel time and layer depth are known. Layer depth (Eq. 3) and associated uncertainty (Eq. 4) when two way travel time and real permittivity are known.

$$\epsilon' = \left(\frac{t \cdot 0.3 \text{ m ns}^{-1}}{2 \cdot z} \right)^2 \quad (1)$$

$$\delta\epsilon = 2 \left(\frac{0.3 \text{ m ns}^{-1}}{2 \cdot z} \right)^2 \cdot t \cdot \delta t \quad (2)$$

$$z = \frac{t \cdot 0.3 \text{ m ns}^{-1}}{2 \cdot \sqrt{\epsilon'}} \quad (3)$$

$$\delta z = \sqrt{\left(\frac{\delta t \cdot (0.3 \text{ m ns}^{-1})}{2 \cdot \sqrt{\epsilon'}} \right)^2 + \left(\frac{(-1) \cdot \delta\epsilon' \cdot t \cdot (0.3 \text{ m ns}^{-1})}{4 \cdot \epsilon'^{(3/2)}} \right)^2} \quad (4)$$

Variables:

ϵ' : real permittivity

$\delta\epsilon'$: real permittivity uncertainty

t : two way travel time (ns)

δt : two way travel time uncertainty (ns)

z : layer depth (m)

δz : layer depth uncertainty (m)

4.2 Fifty meter elevation bin and input mass uncertainty

The following steps were used for uncertainty propagation when calculating the SWE per elevation bin, the glacier average SWE, and the total winter accumulation mass.

1. When calculating the mean two way travel time for each 50 m elevation bin, we also calculated the

standard deviation. We used two standard deviations as the uncertainty in this propagation.

2. Two standard deviations from the mean crossover analysis were added to the uncertainty from step one.

3. Snow density for each segment uses a 9% uncertainty, which is propagated to the column average uncertainty. Eq. 5 is used for the uncertainty on each Seward core, while Eq. 6 is used to calculate the mean between the Seward core mean and the Kaskawulsh core density. The resulting column average uncertainty from both site is 22 kg m^{-3} . We use two times this average for the uncertainty on snow density.

$$\bar{dp} = \frac{1}{N} \sqrt{\sum_N dp_k^2} \quad (5)$$

$$\bar{d\rho} = 0.5 \cdot \sqrt{dx^2 + dy^2} \quad (6)$$

Variables:

$\bar{d\rho}$: final snow density uncertainty; 22 kg m^{-3}

dx : Seward core uncertainty; 38 kg m^{-3}

dy : Kaskawulsh core uncertainty; 21 kg m^{-3}

4. The average snow depth and SWE per elevation bin was then calculated. The average snow depth and snow depth uncertainty per elevation bin was calculated using Eqs. 3 & 4. The snow depth per bin was converted to SWE per bin using Eqs. 7 & 8.

For the real permittivity uncertainty we used two times the standard deviation from the nine equations in Sugiyama and others (2010) for a density of 438 kg m^{-3} which provided $1.86 \pm 0.16 (2\sigma)$. As there is the uncertainty in the snow density ($44 \text{ kg m}^{-3} (2\sigma)$), we also calculated the real permittivity for $394 \text{ kg m}^{-3} (1.77)$ and $482 \text{ kg m}^{-3} (1.95)$, which is a ± 0.09 spread from the real permittivity calculated using the average snow density. We added this value to the uncertainty for the average density providing a resulting in a real permittivity and uncertainty of 1.86 ± 0.25 to be used in the bin calculations.

For the two way travel time uncertainty, we used two times the standard deviation of all two way travel time measurements in each 50 m elevation bin. We added to that the mean crossover analysis result of

62 1.49 ns.

$$s = z \cdot \rho \quad (7)$$

$$\delta s = \sqrt{\left(\frac{\delta z}{z}\right)^2 + \left(\frac{\delta \rho}{\rho}\right)^2} \cdot s \quad (8)$$

63 Variables:

64 s: average SWE per elevation bin (m)

65 δs : average SWE per elevation bin uncertainty (m)

66 z: average snow depth per elevation bin (m)

67 δz : average snow depth per elevation bin uncertainty (m)

68 ρ : snow density to water density ratio (0.438)

69 $\delta \rho$: snow density uncertainty (0.044). 0.044 from the snow density uncertainty divided by the density
70 of water ($44 \text{ kgm}^{-3}/1000 \text{ kgm}^{-3}$).

71 5. When converting from snow depth to SWE, we again used the two standard deviations from the density
72 in the conversion.

$$\bar{S} = \sqrt{\sum_N \delta s^2} \quad (9)$$

73 Variables:

74 $\bar{S}$: glacier average SWE uncertainty (m)

75 δs : SWE uncertainty per elevation bin (m)

76

77 6. The glacier average SWE uncertainty is propagated into the estimated input mass.

$$\bar{m} = \bar{S} \times A \times 10^{-3} \quad (10)$$

78 Variables:

79 $\bar{m}$: glacier input mass (gigatons)

:

80 $\overline{S}$: glacier average SWE uncertainty (m)

81 A: Glacier area from GLIMS (RGI Consortium, 2023) (km)

82 10^{-3} factor: needed for converting m SWE to gigatons. 1 Gt = 10^{12} kg. Converting m to kg uses a
 83 factor of 10^9 .

84 REFERENCES

85 OpenTopography (2021) Copernicus GLO-90 Digital Surface Model (doi: 10.5069/G9028PQB)

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 90 **51**(55), 9–15, ISSN 0260-3055, 1727-5644 (doi: 10.3189/172756410791392745)